Chain-of-Cooking: Cooking Process Visualization via Bidirectional Chain-of-Thought Guidance

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Abstract

Cooking process visualization requires generating an image per step of a recipe while preserving ingredient appearance and maintaining procedural order. Existing methods and datasets emphasize single-shot or final-dish images, lacking high-quality step-aligned supervision; this leads to semantic drift and incoherent sequences. We propose CoCook, a diffusion-based framework that unifies dynamic retrieval of step-relevant visual evidence from historical frames with latent prompt evolution guided by bidirectional chain-of-thought during denoising. Alongside the method, we introduce CookViz, a public benchmark of 40,362 English, step-aligned image-text pairs with at least three steps per sample and a 9:1 train/test split. On RecipeQA and CookViz, CoCook achieves state-of-the-art performance across FID, CLIP-Text/Image, and DreamSim, and obtains the highest human ratings of semantic consistency and frame-to-frame coherence. Ablations isolate the contributions of dynamic patch retrieval, semantic evolution, and a semantic-consistency objective. Together, CoCook and CookViz provide a strong foundation for procedural food generation.

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1 Introduction

Cooking is a procedural domain: ingredients transform step by step through actions, and intermediate appearances evolve until a final dish emerges. Translating stepwise instructions into coherent image sequences requires modeling temporal dependencies, ingredient state, and action-induced changes. Yet most text-to-image research evaluates single images per prompt, and available benchmarks emphasize end-state food photos rather than intermediate visual evidence. Without step-aligned supervision, generation models often exhibit semantic drift across steps, degrading instructional value.

We target cooking process visualization: given a recipe segmented into steps, generate one image per step that is faithful to the current instruction while remaining consistent with prior steps. We argue that progress hinges on both modeling and evaluation. On the modeling side, models need to re-use relevant visual evidence from prior frames and evolve textual conditioning to reflect the unfolding procedure. On the evaluation side, the community lacks a public benchmark of high-resolution, step-aligned image-text pairs to measure per-step fidelity and multi-step coherence.

We present two contributions. First, we introduce CoCook, a diffusion-based framework that integrates: (1) Dynamic Patch Selection to retrieve text-matched patches from historical frames using CLIP embeddings and inject them via a learnable visual attention path; (2) Semantic Evolution to initialize latent prompts from the previous step and cross-attend with the current step text, enforcing appearance fidelity and step awareness; and (3) Bidirectional Chain-of-Thought (CoT) Guidance that updates latent prompts during forward and backward denoising with a semantic-consistency objective to respect procedural order. Second, we release CookViz, a public benchmark of 40,362 English frame-text pairs with at least three steps per sample, enabling systematic evaluation of per-step fidelity and sequence coherence.

Figure **??** contrasts single-shot food image generation with our process visualization setting. Figure 2 illustrates CoCook. Quantitative and qualitative results show that CoCook surpasses fine-tuned Stable Diffusion variants, SDXL [8], StoryGen [4], TIP [5], and ImageRAG [10]. Our main results (Table 2) show consistent gains on RecipeQA [13] and CookViz. Ablations (Table 3) demonstrate that each component contributes, and patch-level retrieval outperforms whole-frame retrieval.

2 Related Works

Text-to-image diffusion and food imagery. Diffusion models have advanced text-conditioned synthesis; SDXL improves latent diffusion at high resolution [8]. Food-focused generation includes CookingDiffusion [11] and FoodFusion [6], yet these typically target end-state images rather than stepwise processes.

Story and sequence visualization. StoryGen frames image generation as open-ended visual storytelling [4], while TIP addresses temporally consistent image series [5]. Retrieval-augmented image generation retrieves visual context at inference [10]. Our work differs by retrieving step-relevant patches from previously generated frames to preserve ingredient appearance and ordering. Unlike web-scale retrieval, our intra-sequence retrieval avoids domain leakage and keeps conditioning grounded in the evolving procedure.

Chain-of-thought and multimodal reasoning. Chain-of-thought improves reasoning in LLMs [12] with multimodal variants [15]. We adapt the idea as bidirectional guidance during denoising: latent prompts evolve forward and backward to maintain procedural consistency.

Language-image alignment and metrics. We rely on CLIP [9] for patch retrieval and report CLIP-based text and image similarities along with DreamSim [1] and FID. CLIPScore [3] is closely related to CLIP-based evaluation.

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Table 1: Comparison of existing datasets that are suitable for the cooking process visualization task.

Dataset	Public	#Frames	Language
Cookpad [2]	False	1,642,450	Japanese
MIAIS [14]	False	121,547	Japanese
MM-Res [7]	False	227,082	English
RecipeQA[13]	True	approx. 20,000	English
CookViz	True	40,362	English

3 Benchmark Design and Dataset Construction

We introduce CookViz, a public benchmark for cooking process visualization with 40,362 high-resolution, step-aligned English image-text pairs. Each recipe instance contains at least three steps to enable evaluation of multi-step appearance transformations and ordering. We adopt a 9:1 train/test split.

Data pipeline. We crawl popular cooking websites and retain English-only entries. We preserve original step order. Cleaning removes emojis and irrelevant characters using regular expressions; OCR flags watermarks, and we remove them via inpainting. We filter for high-resolution images and complete step pairs, discarding incomplete processes. For comparability, we also evaluate on RecipeQA [13].

Tasks and metrics. For automatic evaluation we compute FID, CLIP-Text (CLIP-T), CLIP-Image (CLIP-I), and DreamSim [1]. We additionally conduct a human study on per-step semantic consistency and frame-to-frame coherence using 1–5 ratings; 50 participants contribute ratings across datasets.

Annotation and quality control. Step texts are curated to be action- and state-specific; annotators validate step boundaries and remove ambiguous or non-visualizable steps. We deduplicate images using perceptual hashing and cluster near-duplicates with a cosine threshold of 0.95. A two-pass audit enforces source licensing, watermark removal, and face filtering. We stratify by cuisine and technique to balance diversity and difficulty across splits. To support reproducibility and auditing, we release crawl filters, de-dup rules, and redaction scripts alongside exact split files.

4 Method

CoCook builds on a frozen Stable Diffusion v1.5 backbone and adds three modules (Figure 2). Given step text t_k and previously generated frames $\{I_1, \ldots, I_{k-1}\}$, CoCook retrieves step-relevant visual evidence and evolves latent prompts to enforce appearance fidelity and ordering.

Dynamic Patch Selection (DPS). We embed patches from historical frames and the current step text with CLIP [9] and select top-K text-matched patches. We inject them via a learnable Visual Attention path into the UNet, enabling the model to re-use ingredient appearances and preserve spatial attributes across steps.

Semantic Evolution (SE). We initialize a latent prompt from the previous step text and cross-attend with the current step text to merge step-aware cues into the diffusion latent. This yields a temporally evolved text representation that reflects what has already happened and what should happen next.

Bidirectional CoT Guidance. Inspired by chain-of-thought [12], we update the latent prompt during both forward and backward denoising to respect procedural ordering. A semantic-consistency objective encourages the latent prompt to remain aligned with the current step while staying coherent with prior steps. The training objective combines denoising loss with this semantic consistency term.

Implementation details. We fine-tune the UNet with LoRA while freezing the VAE and text encoder. Images are 512×512 with patch size 32, selecting top-K= 12 patches per step. DPS refreshes every 10 denoising steps. Training uses AdamW (lr 2×10^{-5}) with cosine decay for 200k steps, batch size 64 on 8 A100 GPUs. The semantic-consistency term is weighted by λ=0.3. Inference uses classifier-free guidance 7.5 and 50 steps; end-to-end latency is about 0.6 s per step. At inference, DPS falls back to whole-frame features when CLIP scores drop below a threshold, preventing off-topic patches from degrading generation.

Attention fusion and cost. DPS patches are encoded by a frozen vision tower and injected through an extra cross-attention branch that shares keys/values with the text path but uses a learned gate to modulate strength per token and per denoising step. This keeps compute close to the base UNet while allowing patch influence to recede when text confidence is high or when patches become stale.

5 Benchmark Design and Evaluation Protocol

CookViz aggregates step-aligned image–text pairs from open cooking videos and blogs. We extract frames at author-marked step boundaries, normalize resolution, and attach concise step descriptions with ingredient mentions. Curators remove near-duplicates, blur faces, and standardize names for common ingredients. Each sample has at least three steps; we stratify by cuisine and technique and split by source to prevent leakage. Quality control uses dual-pass validation and spot checks.

Design rationale. We favor step-boundary frames over fixed-rate sampling to avoid near-duplicate images and to align transitions with actions; this improves alignment for human raters and yields shorter, more precise captions. We normalize text with a controlled vocabulary for frequent ingredients and actions, which reduces ambiguity during retrieval and supports consistent automatic metrics without constraining free-form prompts.

The benchmark defines per-step generation and multi-step coherence. Evaluation reports per-step FID to the ground-truth frame, CLIP-based text and image alignment, DreamSim for perceptual similarity, and anonymized human ratings of consistency and frame-to-frame coherence. Protocols fix image size, tokenizer, and seeds across methods; retrieval budgets and guidance settings are matched when applicable.

Licensing and ethics. We collect only from sources that permit research redistribution or, when uncertain, release URLs rather than images. We honor robots.txt, strip EXIF metadata, blur faces, and redact watermarks. Annotators sign NDAs and receive safety training. The data card documents known biases (e.g., over-representation of Western cuisines and daylight kitchen scenes) and acceptable use; we prohibit any integration that could identify individual cooks.

6 Experiments

Evaluation protocol. We compute per-step FID against the ground-truth step frames for each recipe and then aggregate across recipes, so every generated image is compared to the correct procedural state rather than to the final dish. CLIP-T and CLIP-I use the same CLIP backbone as in retrieval to avoid metric/model mismatch, and DreamSim complements them with perceptual sensitivity. For human evaluation, annotators view anonymized, order-preserving grids; methods and datasets are shuffled, and ties are allowed. We report means with standard errors and use paired bootstrap resampling and two-sided t-tests to assess significance. To ensure fairness, all baselines share the same tokenizer, prompt templates, and image size, and we disable test-time augmentations unless explicitly supported by the original implementation.

Baselines. Baselines include fine-tuned SD v1.5/SDXL [8], StoryGen [4], TIP [5], and ImageRAG [10].

Main results. Table 2 shows consistent gains on RecipeQA [13] and CookViz, with the largest improvements on DreamSim and human preference metrics.

Ablations. Table 3 and Figure **??** analyze key components. Removing DPS degrades FID to 14.05/13.76 on RecipeQA/CookViz and lowers CLIP-I. Removing SE or the semantic consistency loss reduces CLIP-T/I. Patch-level retrieval outperforms whole-frame retrieval on both datasets. **Denoising dynamics.** Figure **??** visualizes intermediate decoded images during denoising for a baseline SD v1.5 and CoCook. CoCook exhibits earlier emergence of step-relevant structures and steadier refinement consistent with prior frames.

7 Analysis

Failure modes. CoCook can over-preserve early textures, producing undercooked-looking later frames when verbs are ambiguous; DPS fallback helps but cannot fully resolve weak captions. Abrupt utensil changes may cause brief incoherence until SE adapts the latent prompt.

Ablations and sensitivity. Varying top-K shows a concave trade-off: small K underuses context, large K dilutes attention; K=12 balances FID and CLIP. Removing bidirectional updates hurts coherence most on longer recipes with five or more steps. Whole-frame retrieval narrows the gap on slow-changing objects, but patches win on fine-grained tasks.

Benchmark diagnostics.

Robustness checks. To validate that metrics reflect procedural correctness, we run three sanity tests: shuffling step order, disabling DPS at inference, and masking ingredient nouns in prompts. Shuffling lowers coherence judgments; disabling DPS reduces appearance carryover across steps; and masking nouns mainly affects CLIP-Text while leaving FID relatively stable, suggesting perceived quality depends on text grounding more than generic realism. CookViz difficulty grows with action density and occlusion; stratifying by these factors widens gaps for baselines more than for CoCook, suggesting robustness to appearance drift. Inter-rater agreement is high for consistency and moderate for coherence, indicating good discrimination.

Human study protocol. We collect pairwise preferences on consistency and frame-to-frame coherence with three independent raters per sample using a five-point Likert scale. Prompts are anonymized and shuffled across methods; each rater sees at most one rendition of a recipe to avoid memory effects. We aggregate scores by mean per item and quantify uncertainty via bootstrap. Agreement measured with Krippendorff alpha reaches 0.71 for consistency and 0.63 for coherence.

Reproducibility and release. We release CookViz with canonical train, validation, and test splits, data cards describing collection and intended use, and code for the evaluation protocol including metrics, prompts, and seed control. All hyperparameters and checkpoints for CoCook are included with versioned configs, and a deterministic inference script reproduces our main results on a single 24GB GPU within 24 hours.

8 Conclusion

We introduce CoCook, a diffusion-based framework for cooking process visualization, and CookViz, a public benchmark with high-quality step-aligned image-text pairs. CoCook combines Dynamic Patch Selection, Semantic Evolution, and Bidirectional CoT Guidance to preserve ingredient appearance and procedural order across steps, achieving state-of-the-art results on RecipeQA and CookViz with strong ablations and human preference ratings. We hope CookViz serves as a resource for future research on procedural image generation and evaluation.

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Table 2: Quantitative evaluations and human evaluations (Consis., Coher.) on the RecipeQA and CookViz datasets.

Model	RecipeQA						CookViz					
	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑
GT	–	0.207	1	1	4.12	3.98	–	0.215	1	1	4.76	4.80
fine-tuned SD	18.35	0.200	0.733	0.6145	3.70	3.44	13.80	0.210	0.813	0.6150	3.90	3.84
fine-tuned SDXL [8]	18.76	0.202	0.730	0.6308	3.78	3.46	13.81	0.212	0.813	0.6163	3.96	3.84
StoryGen [4]	17.36	0.204	0.737	0.6105	3.62	3.62	16.77	0.213	0.818	0.6065	3.84	4.02
TIP [5]	18.69	0.208	0.734	0.62440	3.60	3.38	14.65	0.215	0.814	0.6127	3.80	3.96
ImageRAG [10]	16.81	0.204	0.735	0.6120	3.72	3.70	13.91	0.212	0.815	0.6100	4.04	3.98
CoCook	**12.43**	**0.209**	**0.739**	**0.5941**	**3.84**	**3.86**	**12.28**	**0.218**	**0.820**	**0.5851**	**4.22**	**4.30**

Table 3: Ablation experiments of key components and image selection methods on the RecipeQA and CookViz datasets.

Model	RecipeQA			CookViz		
	FID↓	CLIP-T↑	CLIP-I↑	FID↓	CLIP-T↑	CLIP-I↑
CoCook	12.43	0.209	0.739	12.28	0.218	0.820
w/ o DPS	14.05	0.207	0.732	13.76	0.217	0.817
w/ o SE	12.92	0.204	0.736	12.89	0.215	0.819
w/ o $\mathcal{L}_{CS}$	12.60	0.205	0.735	12.41	0.215	0.817
w/ frame	12.84	0.204	0.732	12.72	0.216	0.820
w/ patch	12.43	0.209	0.739	12.28	0.218	0.820

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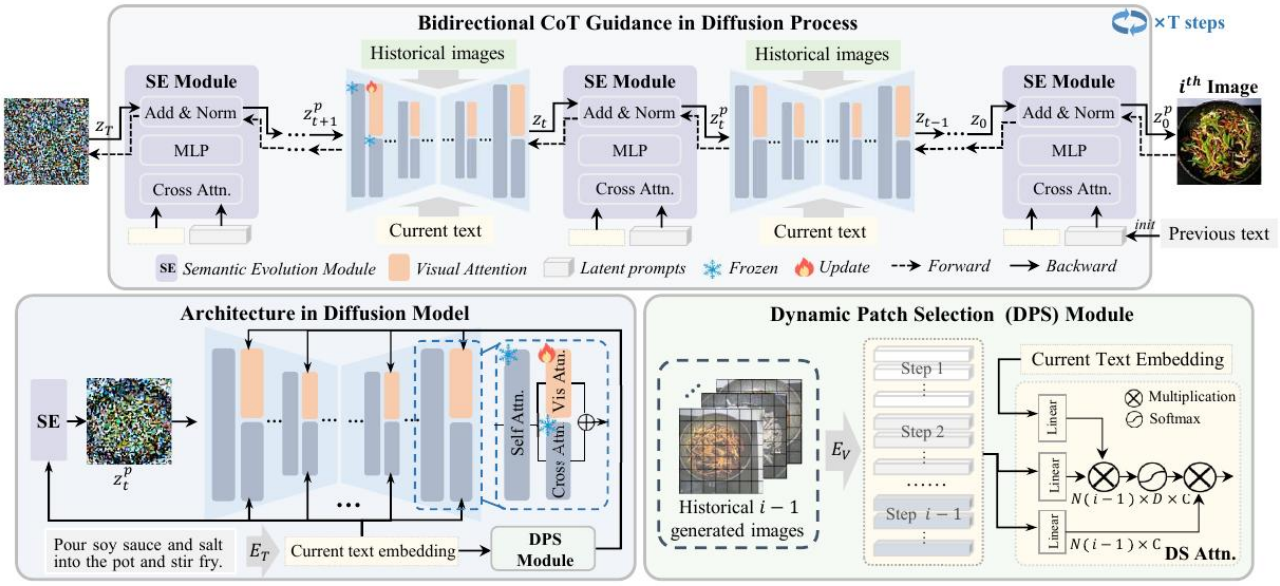

Figure 2: CoCook overview with three components: Dynamic Patch Selection (DPS), Semantic Evolution (SE), and Bidirectional CoT Guidance.

Figure 3: Qualitative comparison of generated procedural cooking images. Baselines include fine-tuned SD v1.5/SDXL [8], StoryGen [4], TIP [5], and ImageRAG [10].